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MELACS7 Boards Design Changes

# MELACS7 Changes for RS485

**Abstract**
MELACS7 Boards update for option with RS485 directly from MELACS7.

| Version | Date | Changed by | Reason for change |
|---|---|---|---|
| A | 2022-01-09 | Rolf Blixt | First version |
| | | | |

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# 1 GENERAL

One extra signal is needed from MELACS7 Board to Extension Board to run RS485 directly from MELACS7. Also making minimal changes on extension Board. “MELACS7” and “MELACS7 processor Board” will be backward compatible after the change except SPI connector JP27 SPIClk(Which easy can be changed back if needed later. ) A SPI connector is used for the data.

With this change we avoid removing functions from Extension Board (Also less modifications of the Extension Board) and are not dependent on other external connectors for the communication between MELACS7 and the Extension Board. (Communications Board not needed for RS485) SPIcommunication not possible parallell when RS485 communication is ongoing.

# 2 CHANGES OVERVIEW

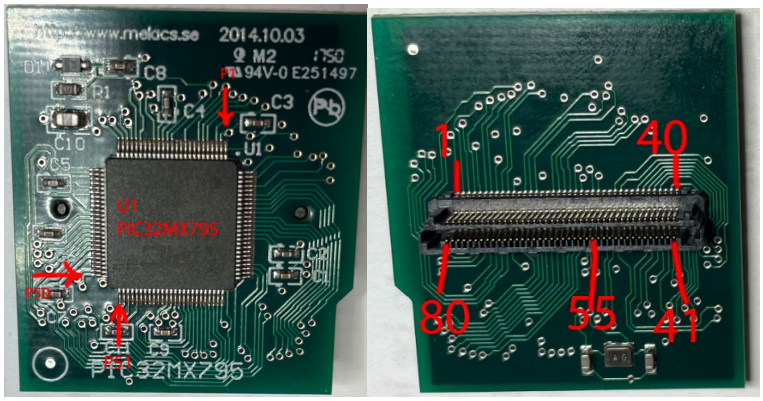
Change 1&2 Wire changes only. (RS485)
**NOTE Change 1 are to be implemented asap for R2A.**

- Change 1: Add signal from MELACS7 Processor Board
  **NOTE Change 2 are to be implemented asap for R2A**
- Change 2: New signal from MELACS7 Processor Board to SPI-connector JP27
- **NOTE Change 3 are to be implemented asap on updated Extension Board rev R5A or later**
- Change 3: Using this new signal previously SPIclk SPI-connector Pin3 Not described in this document. See references.

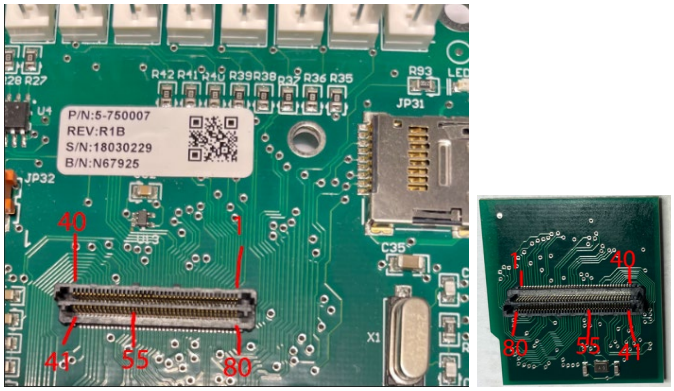
# 3 CHANGES IN DETAIL

## 3.1 CHANGE 1: MELACS7 PROCESSOR BOARD

Adding signal from MELACS7 Processor Board to MELACS7 connector: Processor U1 100pins PIC32MX795L and 80pins SMD-connector J1(Samtec LSHM-140-02.5-F-DV-A-N-K-TR).
**Connect U1P51(USBID/RF3) to J1P55** (J1P55 not used before)
Other unused pins could have been used but U1P51 is in corner so more easy to solder.
(There is a 10u capacitor missing on the PCBA if PCB should be modified later)
Yes U1 and J1 is on different sides of the PCB.



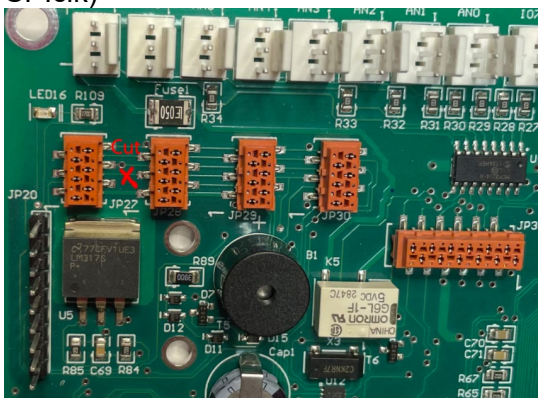

Connect U1P51 to J1P55 on different sides of PCBA see figs above.



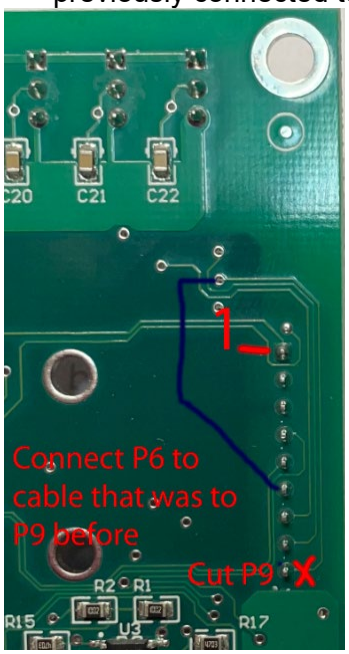

Figs above only for reference because no pin numbering on PCBs.
(J1 pin 49&50 is not used but can be connected to ports on processor Board for future use if you e.g. want to run RS485 without interference of SPI)

## 3.2 CHANGE 2: MELACS7 MAIN BOARD.

U11 (Connected to MEALCS7 Processor Board 80 pins J1)
U11P55(named RB3) on MELACS7 is already connected to programming connector JP20P6 but not used.
6pin connector JP27 is closest to JP20.
**Cut SPI-lead from JP27P3 to JP29P3**.(Makes it disconneted from MPU SPIclk)



- Cut connection from JP20P9 close to pin (marked SPICLK in schema rev7)
- Connect a cable from JP20P6(marked RB3 in schema rev7) to lead previously connected to JP20P9 (Gives connection toJP27P3).




## 4 REFERENCES

PIC32MX795F512L-80 datasheets;
https://www.microchip.com/wwwproducts/en/PIC32MX795F512L

Extension Board update from Rev 03A
MLC-2022-001_Rev_D